

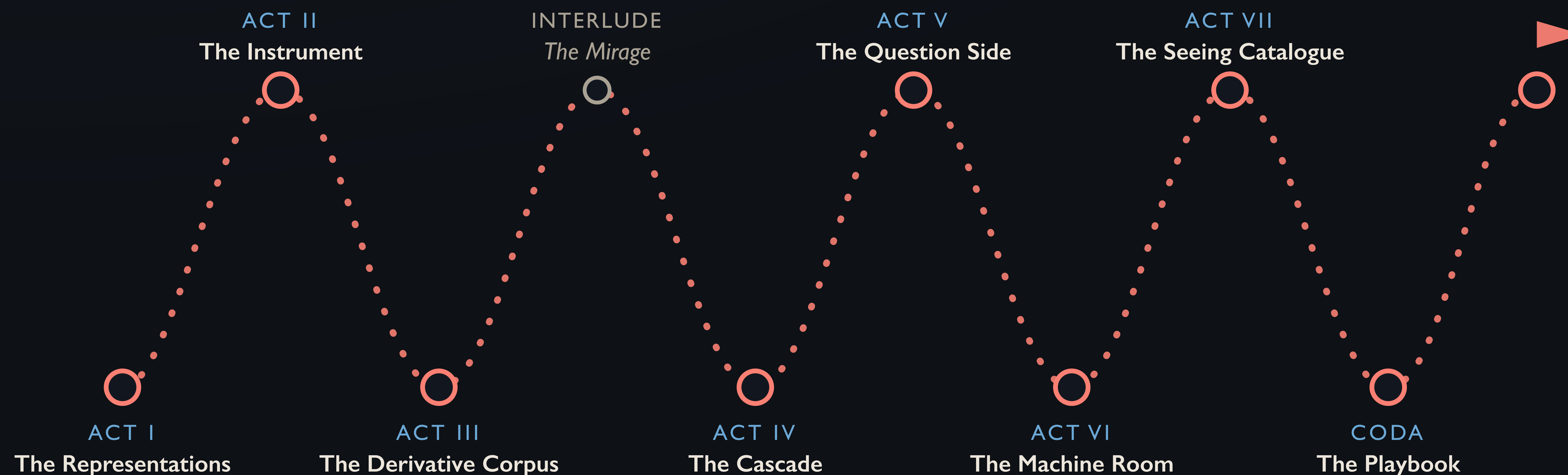
The Library of Many Catalogues

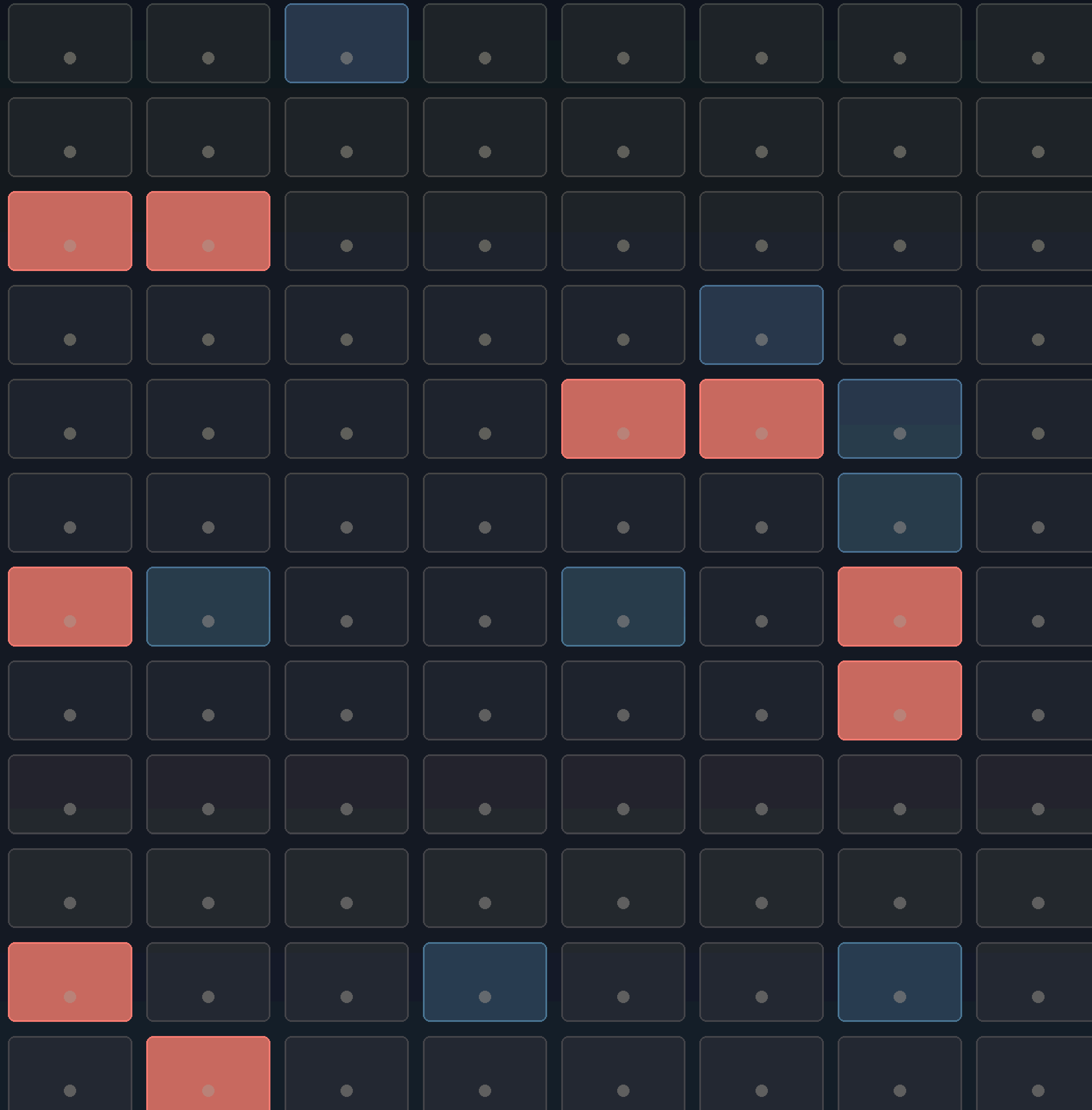
Retrieval Architecture for Enterprise RAG

One corpus is many books. A retrieval system is many catalogues — and a court that decides which card to trust.

THE MAP OF THE DAY

Today's learning journey





PROLOGUE · THE JOURNEY
BEGINS

The Library

A century before RAG, a Belgian lawyer decomposed the world's knowledge onto index cards — and built a portfolio of catalogues by hand.

Otlet's Mundaneum

Otlet and La Fontaine set out to catalogue **everything humanity had published** — the monographic principle: one atomic fact per 3×5 card. **Sixteen million cards** at the peak.

Queries arrived by telegraph; clerks walked the drawers; answers went back by mail — a retrieval service with a latency of days.

Otlet at his desk. (Mundaneum archives.)



One corpus, many catalogues

Many projections of one corpus: by subject, by author, by facet, by the facts themselves. In 1940 the occupiers cleared the halls — sixty-three tonnes destroyed; the surviving drawers reached Mons.

A multi-representation retrieval system, by hand — factoids a century before FactoidWiki. Today we mechanize his library.

The RBU hall, c. 1900.



The recall ceiling

A child loses her marbles in a sandbox; she recovers them with a **bucket and a sieve**. Scoop wide, scoop generously, tolerate sand — the sieve removes sand at leisure.

But **if the scoop misses a marble, no sieve can recover it.**

*Whatever recall you have at the scoop is a **ceiling**. No reranker, no prompt, no bigger model raises it — generation cannot cite what retrieval never surfaced. Only the scoop decides what is caught.*

Portfolio, and court

A retrieval system is not a search over documents. It is:

A portfolio of representations — many purpose-built projections of one corpus.

A court of adjudication — a cascade where cheap judges narrow the field so expensive judges can afford to be careful.

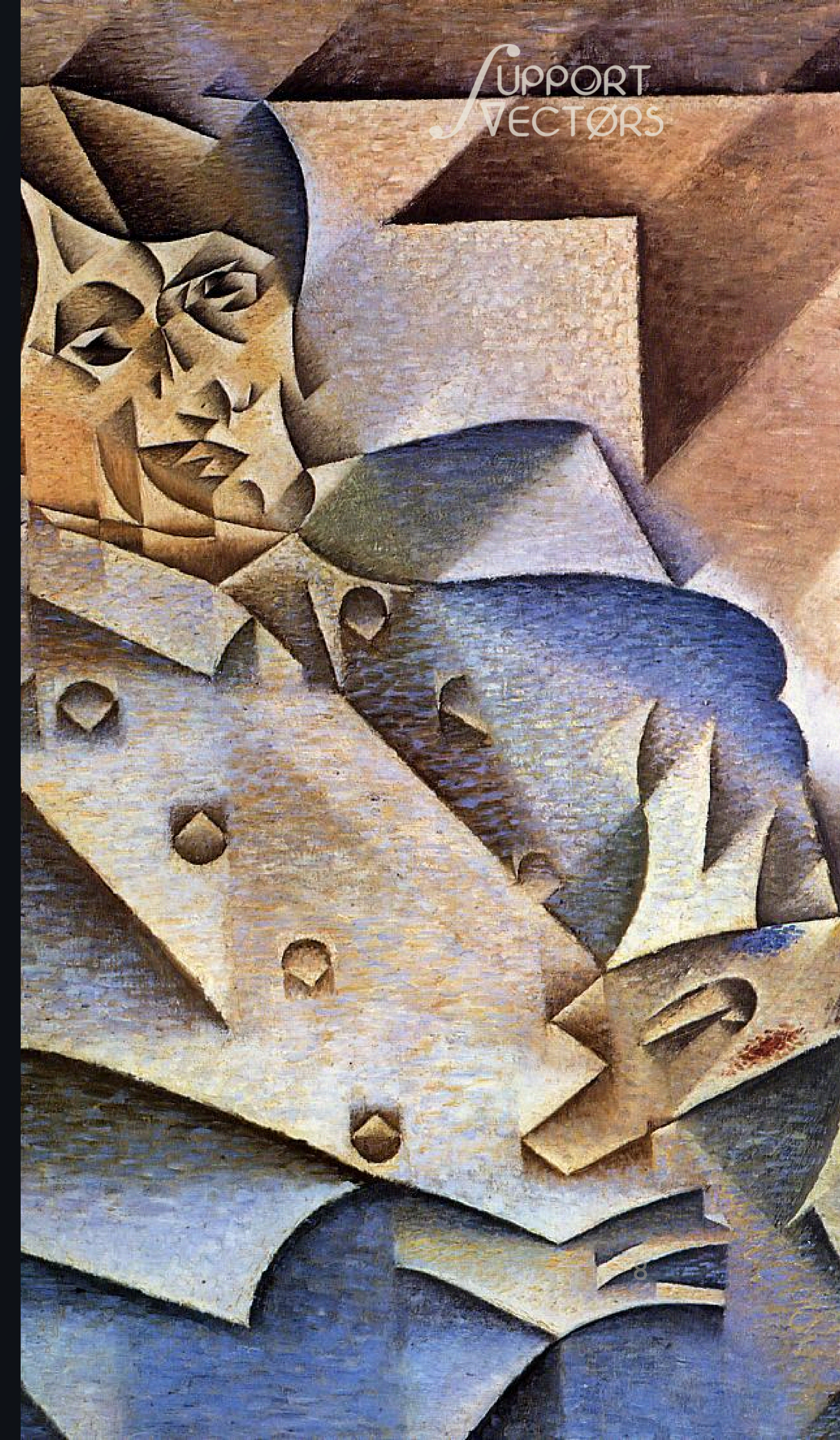
*Architecture is deciding **which representations to manufacture and how to adjudicate among them**. Scale decides how much of each you can afford. Every slide today is a footnote to this thesis.*

The tomographic principle

A CT scanner never photographs the tumor: hundreds of **lossy projections**, each blind alone, **reconstruct** what no single exposure contains. Each index is one projection; **fusion is the reconstruction**.

*The kin: the cubist portrait (right), the compound eye, **Rashomon**. Not the panopticon — one eye claiming total sight. The portfolio refuses that claim.*

Juan Gris, Portrait of Picasso, 1912.
Public domain.



Mein Herr's map

"And then came the grandest idea of all! We actually made a map of the country, on the scale of a mile to the mile!"

"Have you used it much?" I enquired.

"It has never been spread out, yet: the farmers objected — it would cover the whole country, and shut out the sunlight! So we now use the country itself, as its own map, and I assure you it does nearly as well."

*Lewis Carroll, Sylvie and Bruno Concluded, 1893. The perfect map is **no map**: a representation that keeps everything has chosen nothing, and cannot be unfolded. Every useful projection is a decision about what to **attend to** — and what to leave in the dark.*

Borges: the map of the Empire

Borges retold Carroll's joke as history: the Cartographers Guild struck a map of the Empire the size of the Empire, point for point. Later generations found it **useless** and left it to the sun and the winters — tattered ruins in the western deserts.

"On Exactitude in Science," 1946. *The map that omits nothing is abandoned by everyone. A representation earns its use by what it **leaves out**.*



Every map is a choice of what to ignore

The **naval chart** keeps depth soundings, shoals, and lights — and erases every road. The **road map** keeps junctions and distances — and erases the sea floor. The **political map** keeps borders and capitals — no contour lines. The **geological map** keeps strata — no cities at all.

Same country. Four maps. Each one useful because of what it threw away.

A representation is a choice of attention. BM25 attends to the words; the dense lane to the meaning; the factoid to the claim; the summary to the theme. None is the country. The portfolio is the atlas — and the court decides which page to open.

The four query altitudes

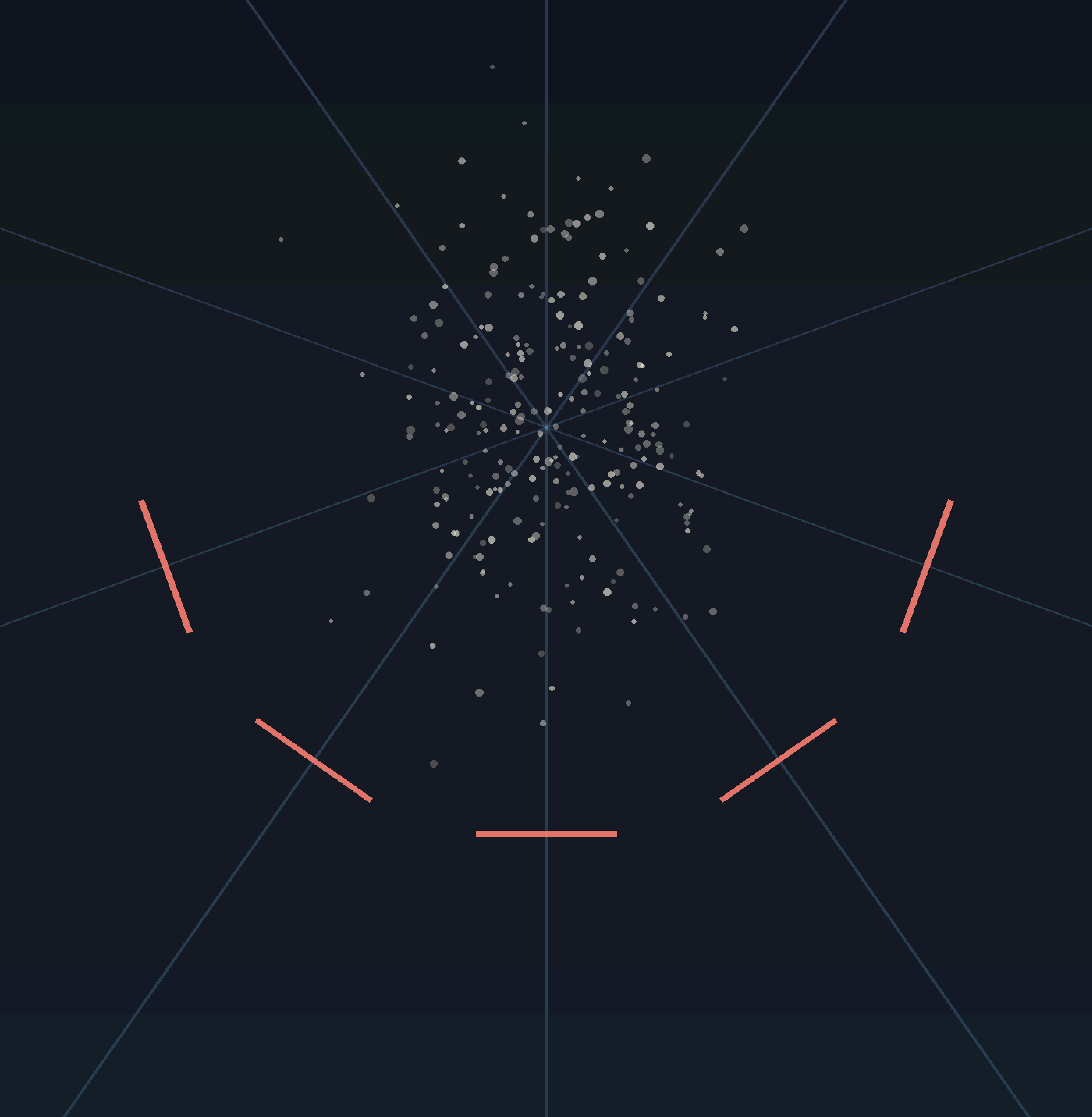
Point — "what is the notice period in clause 12(b)?"

Contextual — "what does **this** section mean for my renewal?"

Thematic — "how has our flood-risk stance evolved?"

Global (sensemaking) — "what are the major regulatory concerns across the portfolio?"

*Misjudging a query's altitude is the most reliable way to build the wrong system. Hold the optics in mind all day: **microscope, zoom lens, telescope** — a well-equipped laboratory owns all three.*



MILESTONE · ACT I OF VII

The Representations

Every index is a lossy projection of the corpus. Architecture begins with choosing which information survives.

ACT I · THE CABINET OPENS

Every representation is a lossy projection

An inverted index keeps the words and discards the meaning. A dense vector keeps the gist and discards the words. A ColBERT matrix keeps both and charges you storage for it.

There is no faithful projection — only **different chosen losses**.

*The design question of Act I is never "which representation is best?" It is: **which information can this query class not afford to lose?** Choose the projection whose loss you can live with.*

BM25 — the words on the page

Robertson and Spärck Jones's lineage: term frequency **saturates** (the 30th mention is not thirty times the relevance), long documents are **normalized**, rare terms carry **more information**.

Fifty years old, and still the baseline every modern method is measured against.

*Its failure mode is exact-match
brittleness: "convolution filters"
in the document, "convolutional
neural networks" in the query —
one inflection, and the match dies.*

***This is not theory. This is
Tuesday.***

BM42 — the launch

July 2024: Qdrant announces **BM42** — replace BM25's term statistics with transformer attention weights. No fine-tuning needed. The launch benchmarks show it beating BM25 and even SPLADE.

The idea is not absurd; it is UniCOIL's learned-reweighting instinct with attention as the teacher.

*The benchmark of choice: **Quora** — a duplicate-question dataset averaging about **1.6 relevant items per query**. Hold that number. It is about to end the story.*

BM42 — the arithmetic that ended it

With ~1.6 relevant items per query, precision@10 **cannot exceed ~0.16** — for any system, ever. The launch numbers were far above the ceiling: the giveaway of a harness bug, not a breakthrough.

Nils Reimers re-ran the evals publicly. The claims collapsed; Qdrant retracted.

*The ceiling is mathematical, not moral — and anyone could have checked it. The lesson is **benchmark hygiene**: extraordinary claims need public evals, and vendor numbers are hypotheses, not facts.*

SPLADE — the words the page means

$$w_j = \max_i \log(1 + \text{ReLU}(z_{ij}))$$

Project each token through the MLM head; keep the sparse max over positions. "The cow jumped over the moon" activates **cattle, bovine, leaped, lunar** — expansion terms the author never typed, stored in an ordinary inverted index.

BM25 sees only the words on the page; SPLADE sees the words the page means.

UniCOIL reweights without expanding — the conservative stop on the same line. The house sparse lane is SPLADE.

Concentration of measure

$$\Pr \left[|\langle u, v \rangle| > \varepsilon \right] \leq 2 e^{-d\varepsilon^2/2}$$

In high dimensions, random unit vectors are **almost surely near-orthogonal** — the sphere's surface huddles at the equator of any pole you pick.

So a high cosine is exponentially unlikely **by chance**.

*Anything found in the polar cap
got there for a reason —
training pulled it there. Signal
survives concentration; noise is
annihilated by it. That is the real
theorem behind semantic search.*

The aperture doctrine

A similarity threshold is a **cone of light** on the sphere: 0.5 opens the aperture to 60°, **0.7 to 45°**, 0.8 to 37°, 0.9 to 26°.

The house recipe is never a bare top- K . It is "**up to 20, all above 0.7.**"

*A bare top-20 is a promise to return twenty things **whether or not twenty relevant things exist**. Forgetting the threshold is how irrelevance leaks into the context window — silently, politely, every query.*

Matryoshka — dimensionality as a cost knob

$$\mathcal{L} = \sum_{k \in \{128, 256, 512, 1024\}} \lambda_k \mathcal{L}_{\text{contrastive}}(e^{(k)})$$

Kusupati 2022: train the loss at **every nested prefix**, and the first 128 dimensions become a usable coarse embedding — a scoop 8× cheaper than the full vector, rescored later at full width.

The compact vector is a coarser net: it catches everything the full vector catches, plus some false positives — a giraffe may occasionally wander into the Arctic cap. At 100M chunks, 128 vs 4096 dims is one server versus a cluster.

ColBERT — every token keeps its vote

$$S(Q, D) = \sum_i \max_j q_i^\top d_j$$

One vector per **token**, matched at query time by MaxSim. The rare term — the part number, the gene name, the **CBTZ** — survives as its own vector and **latches onto its counterpart**, instead of drowning in a pooled average.

The objection was always cost: 128 vectors where one used to live.

*Storage at 10M chunks: **160 GB** vs **1.3 GB**. For years that objection decided the argument. It no longer does.*

The late-interaction cost collapse

ColBERTv2 — residual compression, 16–32× smaller. **PLAID**, then **WARP** — 41× faster than XTR. **MUVERA** — fixed-dimensional encodings make multi-vector search run on ordinary ANN engines.

And **mxbai-edge-colbert**: 17M parameters matching ColBERTv2.

*Within two years, late interaction went from a luxury to a lane you must justify **excluding**. When a cost objection is the entire argument, re-audit the argument **annually** — costs are the fastest-moving part of this field.*

ACT I · THE CHUNK REGAINS ITS MEMORY

The contextualization lineage

Anthropic contextual retrieval (Sept 2024) — an LLM prepends situating context to every chunk: retrieval failures down **67%** with reranking. **Jina late chunking** (same month) — embed the whole document, slice the latents. **voyage-context-3** (July 2025) — beats both, with zero preprocessing.

*Read the lineage as one sentence:
the context is moving into the encoder. What began as prompt-engineered first aid is becoming a property of the embedding model itself. Re-run this bake-off yearly.*

Choosing representations is choosing failure modes

Sparse fails on vocabulary mismatch; dense fails on rare exact terms; both fail differently — their failures are **uncorrelated**, and which query arrives next is aleatoric, irreducible uncertainty.

So take the **union**, never the intersection.

*Hybrid search is not a compromise; it is the only rational strategy under aleatoric uncertainty about query type. A representation set is a declaration: **these are the failure modes we refuse to accept.***

Quiz I — the cap that wasn't full

*Your dense lane is configured as a bare top-20. For one query, exactly **6** chunks in the corpus lie inside the 0.7 cone (45° aperture).*

***(a)** How many of the 20 returned chunks came from the equatorial band? **(b)** What is the best possible precision@20 here?*

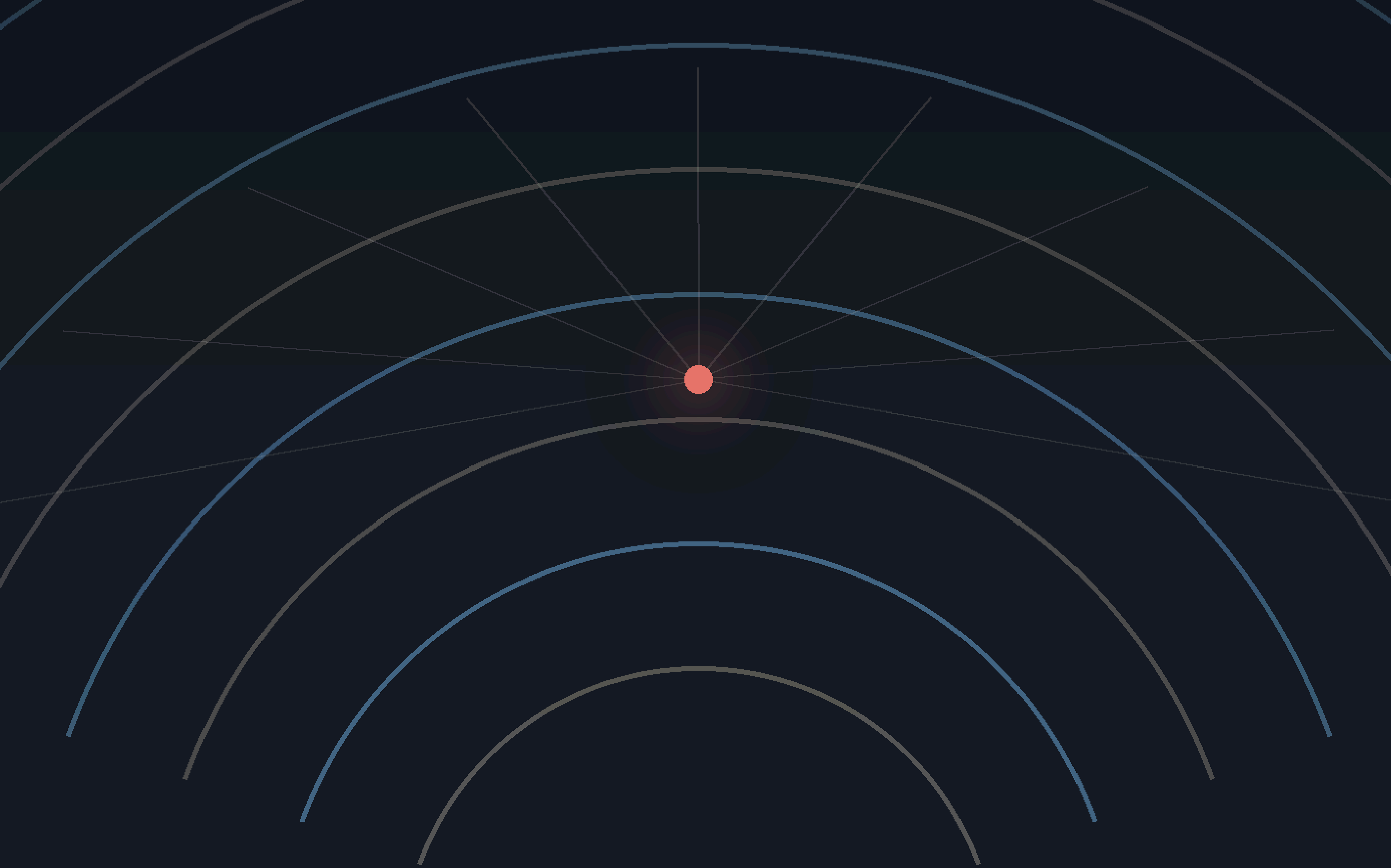
Then say back the house recipe that prevents this.

Fourteen strangers in the context window

(a) $20 - 6 = 14$ — chunks the geometry never endorsed, promoted because the contract said "return twenty."

(b) $6/20 = 0.30$ — a ceiling no retriever can beat, because only six relevant chunks exist.

The recipe: "up to 20, all above 0.7." A top- K without a threshold is a promise to fill the bucket with sand whenever the marbles run out — and the generator downstream cannot tell sand from marble.



MILESTONE · ACT II OF VII

The Trained Instrument

The embedder is not a commodity. It is a scientific instrument — and it must be calibrated to your domain.

Two forces, one geometry

Every contrastive loss is the same physics: an **attractive force** pulling positives together, a **repulsive force** pushing negatives apart. The embedding space your database searches at midnight is the **cooled, frozen residue of that argument**.

The pipeline is only as good as the vectors that enter it.

*A generic MTEB download was trained on someone else's argument, on someone else's data. Fine-tuning re-opens the argument **on your terms** — and everything downstream inherits the improvement for free.*

InfoNCE — the line-up tournament

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{e^{u^\top v_+ / \tau}}{e^{u^\top v_+ / \tau} + \sum_{j \neq +} e^{u^\top v_j / \tau}}$$

The field's default. Differentiate it and the attraction on the positive is $(1 - P_+)/\tau$ — throttled by the model's own confidence. Late in training, when positives are easy, $P_+ \rightarrow 1$ and the learning signal **chokes**.

*The positive sits in its own denominator: attraction and repulsion are **wired in series** when they ought to run in parallel. Easy positives gag the gradient exactly when hard negatives still need pushing.*

DCL — the positive leaves the denominator

$$\mathcal{L}_{\text{DCL}} = -\frac{u^\top v_+}{\tau} + \log \sum_{j \neq +} e^{u^\top v_j / \tau}$$

Yeh et al. 2022: evict the positive term — nothing else. Attraction becomes a **constant** $1/\tau$ that cannot be throttled; repulsion still self-paces over the negatives.

Freedom from the batch-size tax.

*Honesty: the big public embedder recipes still ship InfoNCE variants — at batch 8,192 the coupling barely binds. **We fine-tune at batch 32 on rented GPUs.** DCL is the small-batch house choice, not a universal law.*

The hard-negative curriculum

The house ordering, easy to hard: **random** negatives → **BM25-mined** lexical look-alikes → **in-batch** negatives (free with the loss) → **model-mined hard** negatives from your own retriever's near-misses.

*A mediocre loss with brilliant negatives beats a brilliant loss fed easy ones. The machine, like a student, **learns most from the contrasts that sting** — easy negatives are pushing on doors that are already shut.*

False negatives — and the denoising judge

Mine negatives hard enough and you start mining **unlabeled positives** — genuinely relevant passages your gold set never marked. Train on those and you teach the model to push the truth away.

The RocketQA move: **cross-encoder denoising** — let the judge you already own veto suspect negatives before training.

*And while the cross-encoder is on the bench, distill from it: **margin-MSE** transfers its judgment into the bi-encoder. The expensive judge teaches; the cheap scout learns.*

ACT II · THE INVARIANT THAT GUARDS THE SCOOP

The MRL–DCL invariant

House rule: the DCL loss is applied **inside the Matryoshka summation, at every prefix length** — 128, 256, 512, 1024.

Fine-tune at full width only, and the full-dim metrics **improve** while the 128-d prefix quietly degeometrizes.

*This is **silent scoop rot**: the cheap scoop that feeds your whole cascade rots while every dashboard you watch turns greener. Regression-test recall@500 at 128 dimensions on every fine-tune. No exceptions.*

The 48-hour doctrine

A domain fine-tune of a strong open embedder — curriculum, DCL, MRL-aware — is **about two days of work** on modest GPUs, and it is among the highest-leverage components in the whole architecture.

When **not** to fine-tune: small corpus, generic domain, no eval set to gate the swap.

*Never swap embedders without the yardstick — an embedder upgrade is an **index migration** (Act VI), and an unmeasured migration is a leap of faith with your recall ceiling in hand.*

Quiz 2 — the throttled spring

Late in a fine-tune, a typical positive pair scores $P_+ = 0.98$ under InfoNCE, with $\tau = 0.05$.

***(a)** Compute the InfoNCE attraction magnitude $(1 - P_+)/\tau$ and the DCL attraction $1/\tau$. **(b)** By what factor is InfoNCE's pull throttled?*

Then say back, in one sentence, what "decoupled" decouples.

Fifty times weaker, exactly when it matters

(a) InfoNCE: $0.02/0.05 = 0.4$. DCL:
 $1/0.05 = 20$.

(b) $20/0.4 = 50\times$ — the coupled loss pulls at two percent strength precisely when the remaining hard cases need the gradient most.

*Decoupling separates **attraction** from **repulsion**: the positive leaves the denominator, so the pull toward your positives is constant and cannot be throttled by the model's own confidence. That is the whole surgery.*

Quiz 3 — the silent scoop rot

*After a full-width-only fine-tune, full-dim NDCG@10 rises $0.71 \rightarrow 0.76$. Your nightly job also reports recall@500 **at 128 dimensions**: $0.97 \rightarrow 0.81$. A query has 40 relevant chunks.*

***(a)** How many marbles does the 128-d scoop now miss, on average? **(b)** Can the full-dim rescore recover them?*

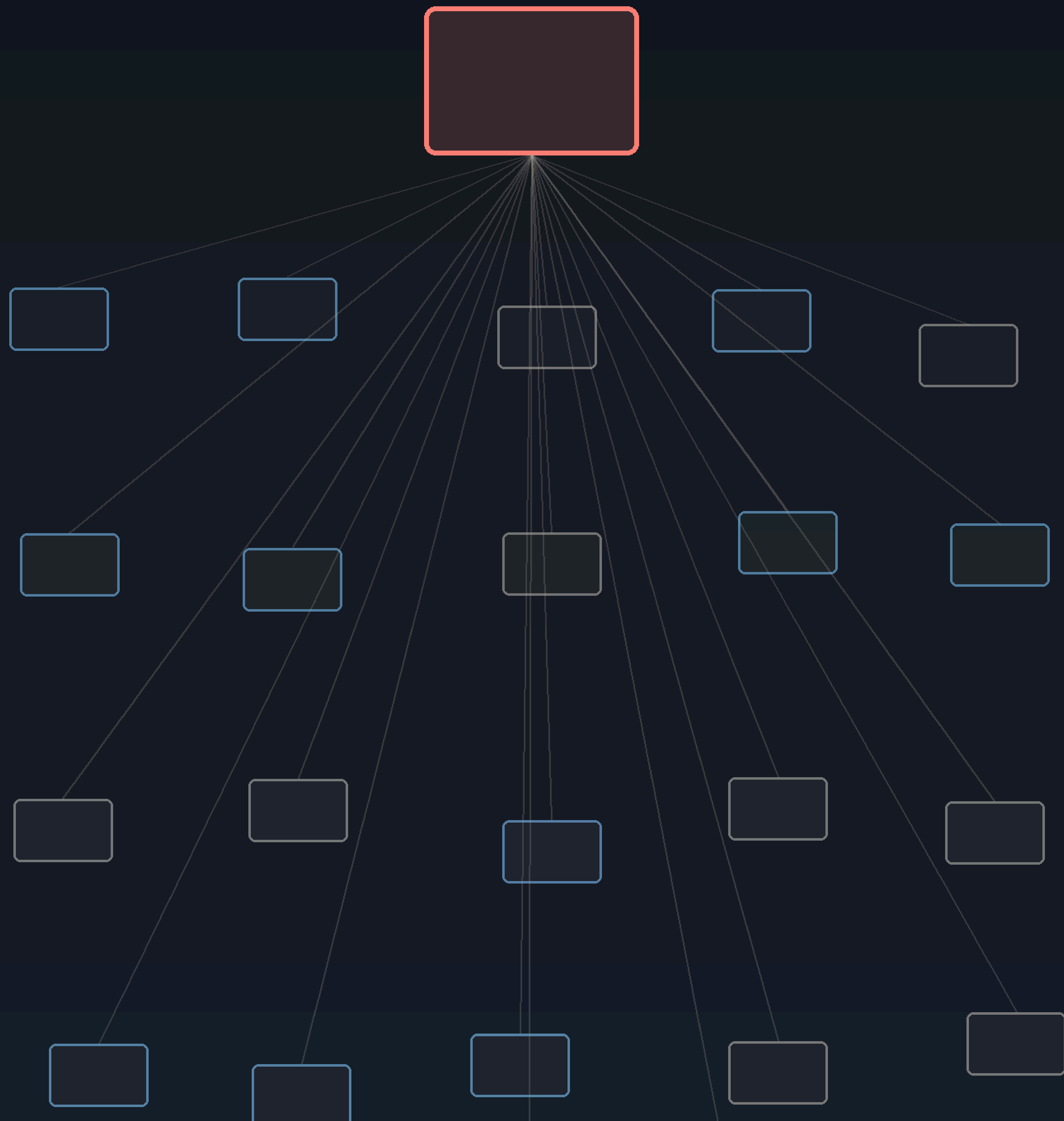
Name the invariant that was violated.

Six marbles, gone before the sieve

(a) $40 \times (0.97 - 0.81) \approx 6.4$ relevant chunks no longer emitted by the scoop.

(b) No. The rescore only reorders what the scoop surfaced — the recall ceiling was lowered upstream, invisibly.

*The violated rule is the **MRL–DCL invariant**: the loss must be applied at every prefix length, and the 128-d prefix must carry its own regression test. Green full-dim dashboards are exactly how this rot stays silent.*



MILESTONE · ACT III OF VII

The Derivative Corpus

Do not wait for the query to
match your documents.

Manufacture documents that
match your queries.

The classmate with forty questions

Applied electrodynamics, IIT. I studied the subject. A soccer-playing classmate visited the seniors and collected the question banks: **roughly forty questions**, recycled across years with small mutations. He memorized beautiful answers, **topped the exam**, took a long shower, and never opened a physics book again.

*I had optimized on learning the subject. He had optimized on **answering the examiner**. A retrieval system faces a stream of questions — and the corpus was written with no examiner in mind at all. **The user is the examiner.***

Retrieve the derivative, generate from the source

Rewrites, factoids, QA pairs, summaries — **searched**. The original passage — **cited and generated from**. Every artifact carries a provenance back-pointer to its source chunk.

The rewrite is the bait; the original is the evidence.

*Learn it once; you will use it everywhere. And the back-pointer is not only the citation — it is the **invalidation key**: summaries are materialized views, and when the source changes, provenance tells you what went stale.*

ACT III · THE GEOMETRY OF THE PROBLEM

The Berlin experiment

Query: "What is the population of Berlin?"

Cosine against the full paragraph: **0.64** — an angle of 50° . Against the extracted factoid: **0.81–0.90**.

At a production threshold of 0.7, the paragraph **containing the answer is silently dropped**.

*A paragraph embedding is the **centroid** of its distinct meanings — near none of them. The answer was there. The geometry hid it. Factoid extraction is the antidote to embedding dilution.*

What the derivative corpus contains

- **Contextualized rewrites** — cure endophora and legalese; the source's understudy, auditioning for the retriever
- **Factoids / propositions** — cure centroid dilution on point queries; Dense X industrialized this: **FactoidWiki, 257M propositions** from 114M sentences
- **QA pairs, 3–8 per chunk** — the only artifact that puts a **query-shaped** object in the index: question as vector, answer in payload
- **Summaries** — section, document, corpus altitudes; thematic queries stop drowning in detail
- **RAPTOR nodes and community summaries** — the zoom lens and the telescope, next two slides
- **Governed concepts** — last week's guest; its seat here in four slides

*Each artifact exists to prevent a **specific failure**.*

Factoids sacrifice context for precision; summaries sacrifice precision for context. Neither is superior — that is the complementarity.

RAPTOR — the query finds its own level

Cluster, summarize, **recurse** — a discovered hierarchy, 1.3–1.5× the leaves. Retrieve over the **collapsed tree**: the query lands at its own altitude.

*Bacon's tree (right) was **decreed**;
RAPTOR **grows** one from the geometry.
Warning: garbage clusters yield confident
garbage summaries — RAPTOR
amplifies embedder quality, both ways.*



GraphRAG — asking the city, not the shelves

Extract entities and relationships, detect **communities** (Leiden), summarize each community. "Major regulatory concerns across 500 earnings calls" exists in **no document** — it lives in the structure.

Embeddings say: these passages sound alike.
Triplets say: these passages are about the same things.

*Some questions can only be answered by the **structure** of a corpus, never by any passage within it. The community summary is the highest-value derivative artifact we have met — a mini-textbook written by the whole corpus.*

The sidecar doctrine, and the cost cliff

Sensemaking is **5–15%** of enterprise queries; in our production routing, **1–2%** of traffic actually enters the graph sidecar. One intern once wired extraction to a frontier API instead of the local cluster: **\$1,000 overnight**, for two Dostoevsky novels.

*The 2024–26 correction:
LazyGraphRAG indexes at
~0.1% of full cost; **HippoRAG 2**
keeps factoid parity while winning
multi-hop. Graph RAG is now a
**specialized instrument, not a
default** — a sidecar, never the
main road.*

The archaeology of top-K

Trace every winner in your top-K back through its provenance: raw chunk, factoid, QA decoy, rewrite, summary node?

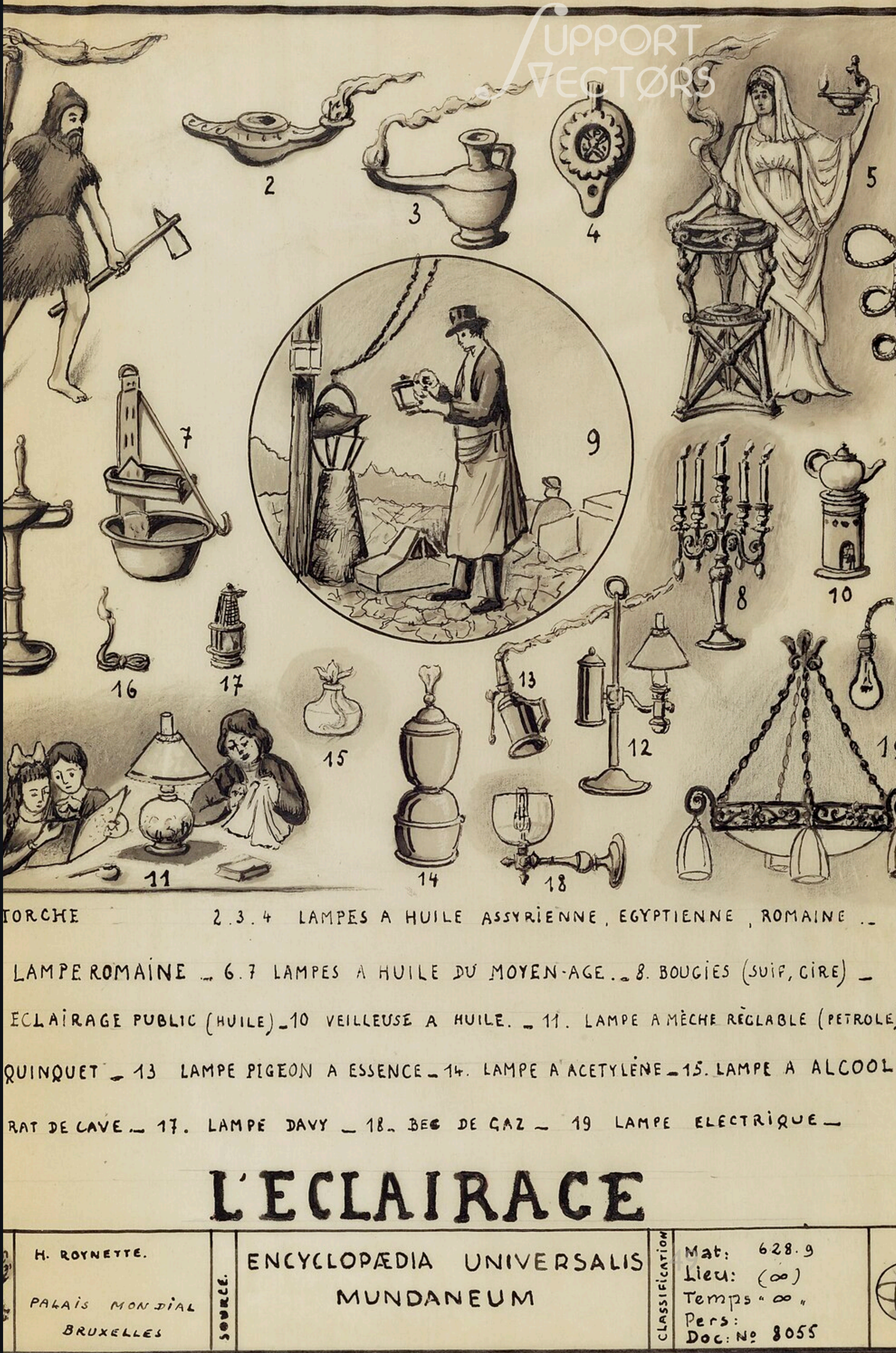
The reproducible finding: **most of your top results, most of the time, are not raw chunks.**

This is the single strongest argument for the multi-representation architecture — no new hardware, no bigger model, spectacularly better retrieval. Run it offline as an audit; keep it running online as telemetry.

Knowledge with papers

Last Saturday, knowledge got its papers. Today OKF takes its seat here: the **governed concept**, the seventh artifact. Its corpse: the **authority failure** — text no one currently stands behind.

*Chunks answer what is similar; concepts answer who says so, as of when — as queryable fields. The plate's cartouche (right): **frontmatter in ink**, c. 1930.*



Extract → review → merge

Every other artifact is derived by pipeline. This one's derivation passes through a **human review gate** — extraction lands as a pull request; **merge is the promotion event**. Repair is a one-line diff, not a re-ingestion.

Its seat: the **top rung** of the ladder — the governed core only (glossary, metrics, policies, runbooks), where an owner will actually review.

*An unreviewed bundle is **chunks wearing a suit** — worse than chunks, because the suit reads as authority. And beware the phantom cluster's descendant: the **phantom concept**, a fluent interpolation wearing the review gate's own seal. Fidelity review, not form review.*

Quiz 4 — the examiner's forty questions

*A compliance query scores **0.64** against the source paragraph and **0.86** against its extracted factoid. The dense lane runs "up to 20, all above **0.7**."*

- (a) Which representation gets retrieved?*
- (b) What text does the generator receive, and under what doctrine?*

Say back why we manufacture the factoid instead of lowering the threshold.

The factoid is found; the source testifies

(a) Only the factoid — $0.86 > 0.7 > 0.64$; the paragraph never crosses the aperture.

(b) The **source paragraph**, resolved via the provenance back-pointer: retrieve the derivative, generate from the source.

*Lowering the threshold to 0.6 widens the cone to $\sim 53^\circ$ **for every query** — admitting the equatorial crowd to fix one miss. The factoid moves **this** answer into the cap instead. Fix the geometry, not the gate.*

Quiz 5 — the telescope and the wristwatch

*Your system serves 100,000 queries/month. Five percent are sensemaking, routed to the graph sidecar at **\$0.20–0.60 per query**.*

***(a)** Monthly sidecar bill? **(b)** The router breaks and routes everything through the graph — new bill?*

Say back the sidecar doctrine, and name the anti-pattern in (b).

\$1,000 well spent — or \$60,000 on fire

(a) $5,000 \times \$0.20\text{--}0.60 = \$1,000\text{--}3,000$ —
buying answers no other instrument can
produce.

(b) $100,000 \times \$0.20\text{--}0.60 = \$20,000\text{--}60,000$
— mostly spent answering point queries.

*The doctrine: GraphRAG is a **sidecar** for the small fraction of queries that need structure — disproportionately the ones executives ask. The anti-pattern is the point query through the graph: **using a telescope to read your own wristwatch.***



INTERLUDE · BETWEEN THE ACTS

The Long-Context Mirage

A million tokens of context
— and still the question: who
reads the library?

NoLiMa, and context rot

Million-token windows tempt a simple heresy: just put the corpus in the prompt. Then remove the lexical overlap between needle and question — **NoLiMa** — and **11 of 12 frontier models fall below 50%** of their short-context performance by 32K tokens.

*Chroma's context rot report ran eighteen models — GPT-4.1, Claude 4, Gemini — and found degradation **well inside the advertised window**. Lost-in-the-middle is mitigated in new models, not solved. Test cures with NoLiMa-style probes, not marketing decks.*

The meter is running

Attention re-reads every token, **every query** — you pay $O(n)$ per token served, at tens of seconds of latency. Retrieval pays $O(\log n)$ -ish per query against a pre-built index.

Corpus-in-the-prompt versus a tuned cascade: roughly a **1,250-fold** per-query cost gap.

*The number will age; the asymmetry will not. Indexing is a one-time capital cost, amortized over every future query. Prompt-stuffing is an **operating cost you pay again, in full, forever.***

INTERLUDE · THE HONEST SYNTHESIS

You do not read the library

You do not read the library to answer a question; **you consult its catalogues.**

What long context genuinely bought us: the retrieved-evidence window grew from 4K toward 100K — bigger top-K, whole-document stuffing at small scale, rerankers that read 32K.

*Long context raised the ceiling on **what retrieval may return** — not on whether retrieval is needed. That is the mid-2026 consensus, and it is a hybrid's consensus, not a partisan's.*

THE LIBRARY OF MANY CATALOGUES · SUPPORTVECTORS

Quiz 6 — the million-token bid

*A vendor proposes skipping retrieval:
stuff ~800K corpus tokens into every
prompt at \$2.50 per million input tokens.*

*Your cascade currently costs **\$0.002–
0.005 per query.***

***(a)** Vendor's per-query cost, and the ratio
to yours? **(b)** Even if cost were zero,
which measured phenomenon still argues
against the bid?*

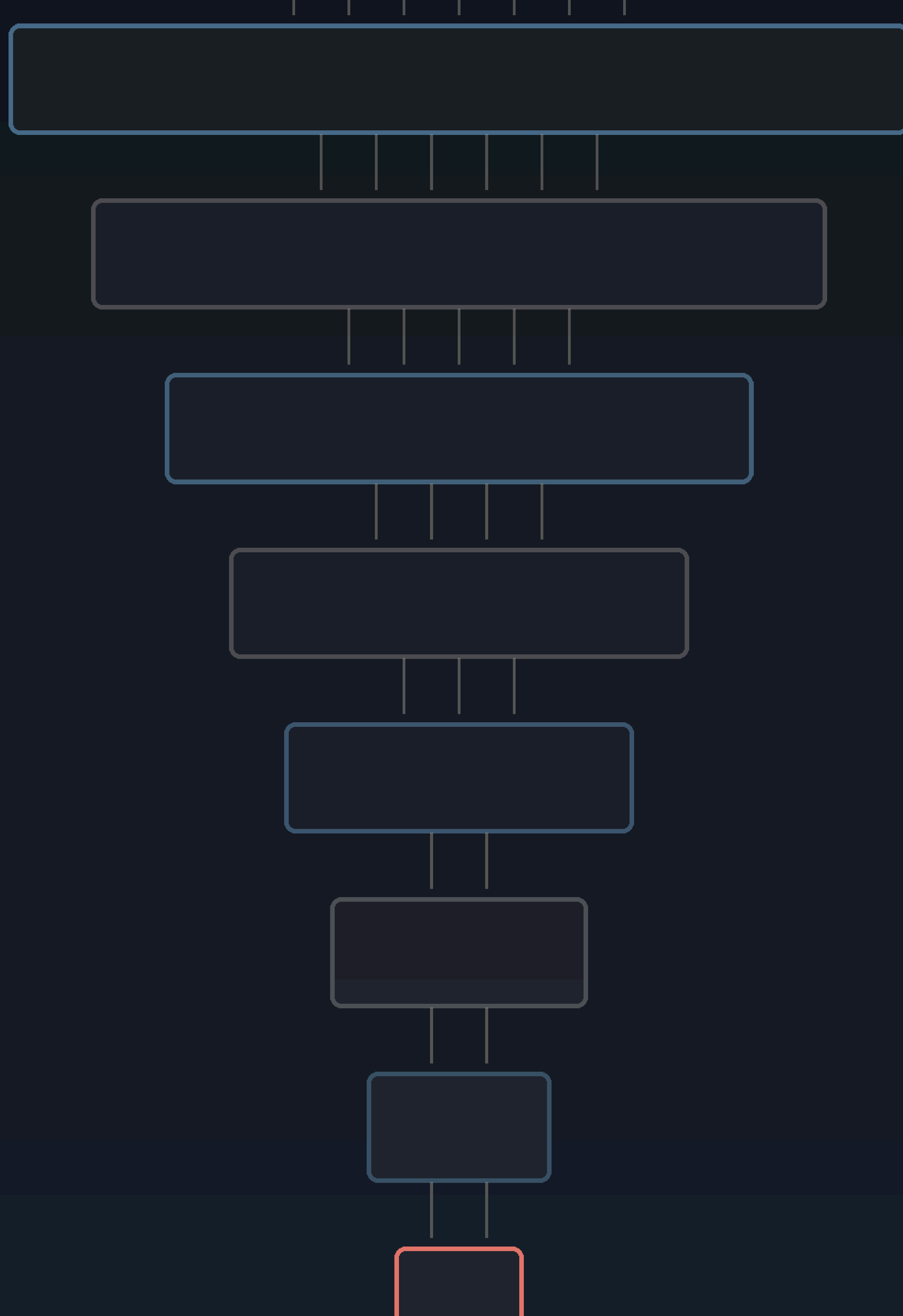
***Say back the library argument in one
sentence.***

Two dollars a question, and the model stops reading

(a) $0.8 \times \$2.50 = \2.00 per query — **400–1,000×** the cascade, in line with the $\sim 1,250\times$ reference gap.

(b) **Context rot:** NoLiMa shows most models losing half their performance by 32K once lexical overlap is gone — long before 800K.

*You do not read the library to answer a question; **you consult its catalogues.** The window is where retrieved evidence goes — it was never a substitute for deciding what deserves to be there.*



MILESTONE · ACT IV OF VII

The Cascade

Cheap judges narrow the field so expensive judges can afford to be careful.

The judicial ladder

District court — the scoops: SPLADE and Matryoshka-128, fast and lenient; recall is sacred. **Appeals** — full-dim rescore. **High court** — ColBERT MaxSim. **Supreme court** — the cross-encoder: the judge who sees little and thinks hard.

Design invariant: **compute per stage stays roughly constant.**

*As candidates fall, **information per candidate rises** — one vector, then many vectors, then the full text read side by side with the query. Rank is cheap; verdicts are expensive.*

Reciprocal rank fusion

$$\text{RRF}(d) = \sum_r \frac{1}{k + \text{rank}_r(d)}, \quad k = 60$$

Cormack, Clarke and Büttcher, 2009. Sum reciprocal ranks across lanes; $k = 60$ remains the default constant in essentially every engine.

It rewards **consensus** across lanes — and still rewards strong **conviction** from a single one.

*RRF is **ordinal, not cardinal**: it reads positions, never scores. That is not a limitation. It is the entire reason it works — as the next slide, and one quiz, will make painful.*

Incommensurable scales

A SPLADE score of 12.4, a cosine of 0.83, a MaxSim of 41.7. These are **different physical quantities** — different ranges, different distributions per query, drifting with every re-index and model swap.

Raw-score normalization across lanes is **technical debt** wearing a formula.

*Every normalization scheme freezes an assumption about score distributions that tomorrow's re-index silently breaks. Ranks are the one currency every lane already speaks — fuse in ranks, then **learn weights** on top.*

The staircase you cannot differentiate

Once lanes have **weights** w_r and **emission depths** k_r , someone must choose them. But NDCG as a function of weights is **piecewise-constant** — a staircase: nudge a weight, no rank flips, the metric does not move; nudge again, a flip, a jump.

No gradient. Nothing to descend.

*Grid search over a dozen lanes is combinatorially hopeless, and hill-climbing on a staircase walks in circles. The house answer is to stop optimizing the staircase — and optimize a **surrogate** of it.*

The surrogate-surface method

Sample weight vectors and per-lane depths $\theta = (w_1 \dots w_R, k_1 \dots k_R)$ at random. **Evaluate** $L(\theta) = 1 - \text{NDCG@10}$ on the train split. **Fit** a linear surface $L \approx a + b^\top \theta$. **Walk** downhill on the plane; sample again around the landing zone. **Confirm on held-out.**

*A poor man's Bayesian optimization — today you would hand the same mixed space to **Optuna/TPE**. The permanent warning: with a dozen lanes and a few hundred queries, the surrogate will happily overfit the sample. **Held-out confirmation is not optional.***

Parent-level dedup — then MMR

After fusion, the docket holds the same source **photocopied**: its factoid, its QA decoy, its rewrite, its summary — all children of one chunk. Cosine dedup and MMR compare **surface forms**; a question and its paragraph do not look alike.

So collapse by **source-chunk id** first, keeping the best-ranked representative.

MMR diversifies surface forms; parent-level dedup collapses identities. You need both, and dedup goes first — it is also the archaeology of top-K, performed at runtime on every query.

The decoy's job ends at retrieval

The QA decoy was **manufactured to resemble questions** — so a cross-encoder scoring query-vs-decoy certifies only that the bait resembles the hook. A verdict on the decoy is void.

Every derivative candidate is **resolved to its source text** before reranking. The cross-encoder judges evidence, never bait.

One honest exception: a summary retrieved for a thematic query is genuinely the evidence at its altitude — rerank it as itself.

Granularity needs quotas or routing; a passage-relevance judge misjudges altitude.

Cross-encoder economics, 2026

Batched on a modern GPU: about **2.5 ms per candidate**. Cohere Rerank 4 Fast: ~**447 ms** on a top-100 docket. Voyage rerank-2.5: **+7.9 points** over the prior generation, a 32K context, instruction-following.

Reranking 100 is routine. Reranking 100,000 is the fantasy the cascade exists to avoid.

Listwise LLM rerankers — RankGPT, RankZephyr — cost 10–100× and, on tuned pipelines, often lose to a fine-tuned cross-encoder. Pay for careful judges, not expensive theater.

Calibration at the exit

The moment anything downstream **thresholds** on the final score — a "sufficient evidence" gate, an abstention rule — you are betting the number means what it says. Raw reranker logits do not.

Platt scaling or isotonic regression on a held-out set turns scores into probabilities you may threshold.

Decisions first, probabilities only if calibrated. An uncalibrated 0.92 waving a hallucinated answer through the grounding gate is the cascade's last and quietest failure mode.

ACT IV · THE WHOLE COURT, WITH NUMBERS

The reference cascade

STAGE	COURT	INSTRUMENT	EMITS
1a	district	SPLADE sparse scoop	100
1b	district	Matryoshka-128 dense scoop	500
2	appeals	full-dim rescore of 1b	250
3	high	CoBERT MaxSim filter	100
4	—	RRF fusion of lanes	~150
5	—	parent-level dedup, then MMR	~100
6	supreme	cross-encoder on source text	20–30

*The numbers are illustrative; **the architecture is the invariant, the numbers are the variables** — retuned by the surrogate surface against your own gold set. Output: 20–30 seats in the generator’s context window.*

Quiz 7 — the incommensurable scoreboard

*Three top-hit scores: SPLADE **12.4**,
cosine **0.83**, MaxSim **41.7**. Which
normalization lets you sum them into one
fused score?*

*(a) min–max to $[0, 1]$ (b) z-score per lane
(c) divide by lane max (d) softmax per
lane*

Decide — and defend your choice.

None of the above — use ranks

Every option performs **cardinal surgery on incommensurable quantities**: (a) and (c) are hostage to one outlier, (b) assumes stable per-lane distributions, (d) invents a probability that re-indexing breaks tomorrow.

The correct move discards the scores:

$$\text{RRF} = \sum_r 1/(60 + \text{rank}_r).$$

*If you rejected all four options — good. **The option to reject every option is always on the ballot.** A system, like a student, must be able to say “none of these is right”; that refusal is negative rejection, practiced live.*

Quiz 8 — the decoy that testified

A QA decoy — "What is the notice period for termination?" — wins retrieval, answer and source-chunk id in its payload. What goes to the cross-encoder against the user's query?

(a) the decoy question (b) the payload answer (c) decoy + answer concatenated (d) the contextualized rewrite of the clause

Choose, and justify with the doctrine.

Again none — the source takes the stand

Every option reranks a **derivative**. (a) certifies bait resembles hook; (b) and (c) grade synthetic text the generator will never cite; (d) auditions the understudy in the actual play.

Follow the provenance pointer: rerank the **source clause** itself.

*Caught twice in a row —
wonderfully. Two traps make the
reflex: **when every offered
answer reranks a stand-in,
reject the slate.** The decoy's job
ends at retrieval; only evidence
testifies before the supreme court.*

Quiz 9 — the cascade by the numbers

*Two lanes, RRF with $k = 60$. Document **A**: rank 2 in the sparse lane, rank 40 in the dense lane. Document **B**: rank 8 in both.*

Compute both RRF scores. Who wins?

Then say back what RRF just rewarded — and what it deliberately ignored.

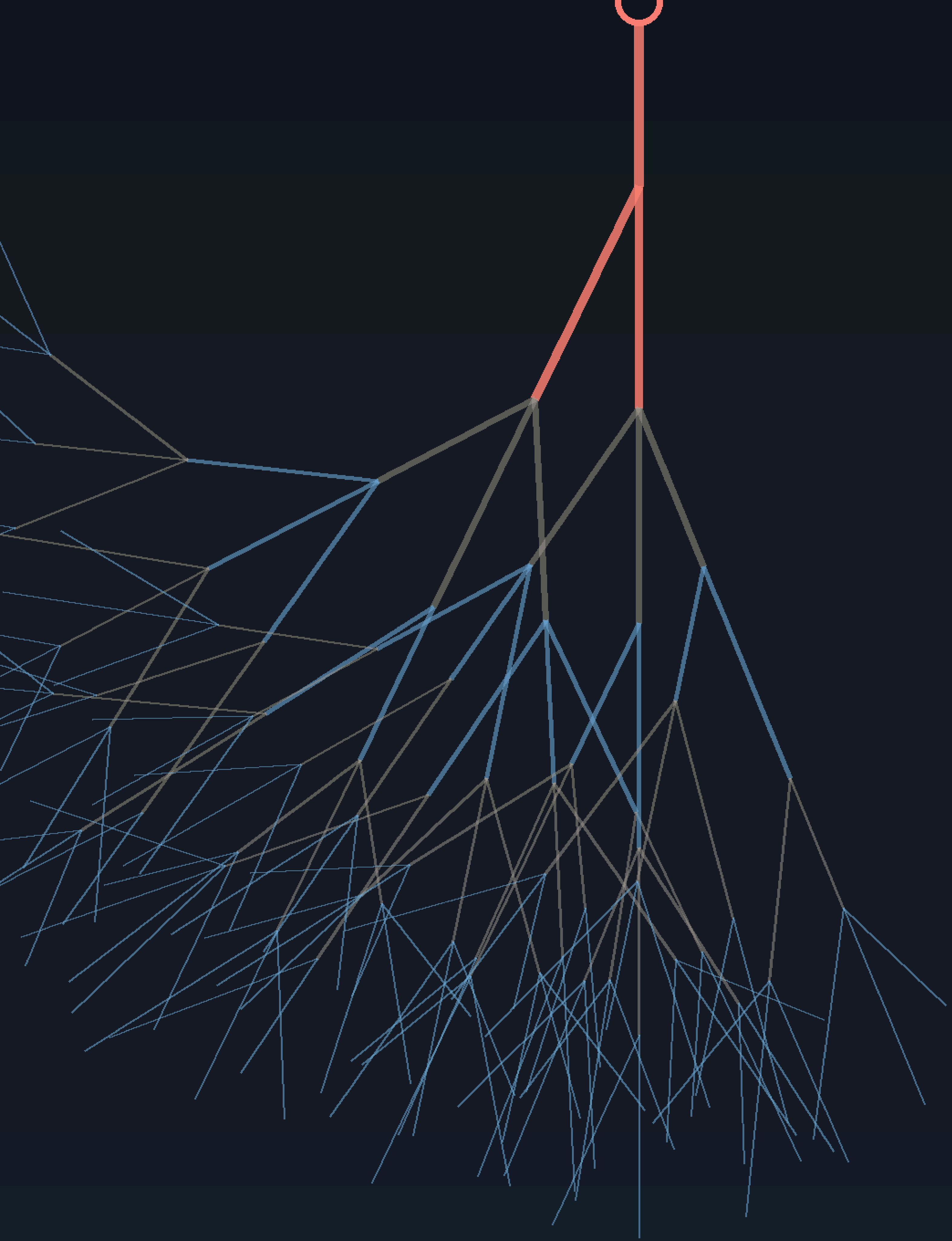
Consensus edges out conviction — this time

$$A : \frac{1}{62} + \frac{1}{100} = 0.0161 + 0.0100 = \mathbf{0.0261}$$

$$B : \frac{1}{68} + \frac{1}{68} = \mathbf{0.0294}$$

B wins — two solid endorsements beat one brilliant showing plus one shrug.

*RRF rewarded **cross-lane consensus** while still letting A's rank-2 conviction nearly close the gap — and it ignored every raw score along the way. Ordinal fusion, doing exactly what it promised.*



MILESTONE · ACT V OF VII

The Question Side

Half the architecture stands
on the query's side of the
glass.

Six pathologies, six verbs

- **Surface errors** — typos degrade subword embeddings → **correct**
- **Context starvation** — the follow-up equidistant from everything, close to nothing → **inject**
- **Jargon opacity** — "CCAR" goes searching for its own meaning → **expand**
- **Compositional complexity** — three questions wearing one trench coat → **decompose**
- **Semantic vagueness** — a wish, not a query → **sharpen**, or HyDE
- **Temporal ambiguity** — "the latest policy": as of when? → **resolve**

*Each pathology gets a verb, each verb a stage. The pipeline is complete when **every pathology has a stage that addresses it.***

The transformation cascade

Stages ordered **cheap to expensive**, with short-circuiting: spell → context injection → acronym expansion → rewrite/expand → decomposition → HyDE as last resort.

Two passes around the guardrails: deterministic fixes **before**, LLM-assisted rewriting **after** — never amplify adversarial content pre-screening.

*The output is the **canonized query** — and it, not the raw string, is the semantic-cache key. "WFH policy" and "remote work policy" must converge **before** the cache, or the second user misses.*

HyDE — the document-shaped probe

Ask an LLM to **answer** the question first; embed the **fabricated answer**; search with that. The hallucinated answer lands in Seattle — the real documents are in the nearby suburbs. Wrong in the particulars, right in the register.

*HyDE crosses the query–document register gap by generating the missing side — the **online twin of the offline QA pair**: one principle, two timings. It is also the costliest stage and playing with fire on factual precision: last resort, not default.*

The semantic cache — E and τ

$$\text{hit} \iff \max_i \text{sim}(E(q), e_i) > \tau$$

One embedder E , one threshold τ — every cache failure you will ever debug traces to one of them. Two tiers: above τ_{full} , serve; between τ_{partial} and τ_{full} , **do not serve** — use as a retrieval hint.

τ is an aperture; **never KNN**
without a threshold — the type-1 vs type-2 diabetes collision is one careless cone away. Production hit rates run 20–45% (folklore — measure your own), and invalidation must track index versions.

Routing, and the head/tail split

Classify by **altitude** (point / thematic / global) and **modality** — structured questions go to the text-to-SQL sidecar, because you cannot retrieve your way out of a schema problem.

Instacart's split: the **head** precomputed and cached (~98% precision), the **tail** served by a distilled 8B model at ~300 ms.

*Routing is also the maturity path for fusion: one global weight vector is a compromise across altitudes. The mature system classifies first, then **reweights its lanes per query class**.*

The agentic turn

Search-R1 trains the LLM by RL to interleave reasoning with search — **+41%** over prompting. On **BRIGHT**, the reasoning-retrieval benchmark, launch SOTA was **~18** nDCG@10 in 2024; the leaderboard reads **66.9** by 2026.

Retrieval learned to reason — marketing in 2023, a measured fact now.

*And yet: the loop pays its latency bill on every hop. For most enterprise traffic — point queries, cached heads — the **single-shot cascade still wins**. Route the hard residue to the agent; do not make every query pay agent prices.*

When not to retrieve at all

Chitchat needs no index. A cache hit needs no cascade. A question the model answers with calibrated closed-book confidence needs no evidence run.

The best retrieval architecture includes a principled way to **stay home**.

*This is the abstention kinship: the system that knows when it does not know is kin to the system that knows when it **need not look**.*

Both are threshold decisions — and both demand calibration first.

Quiz 10 — the fortune teller's suburb

Cache thresholds: $\tau_{\text{full}} = 0.92$,
 $\tau_{\text{partial}} = 0.80$. Three canonized queries
arrive with nearest-neighbor similarities
0.95, 0.86, 0.74.

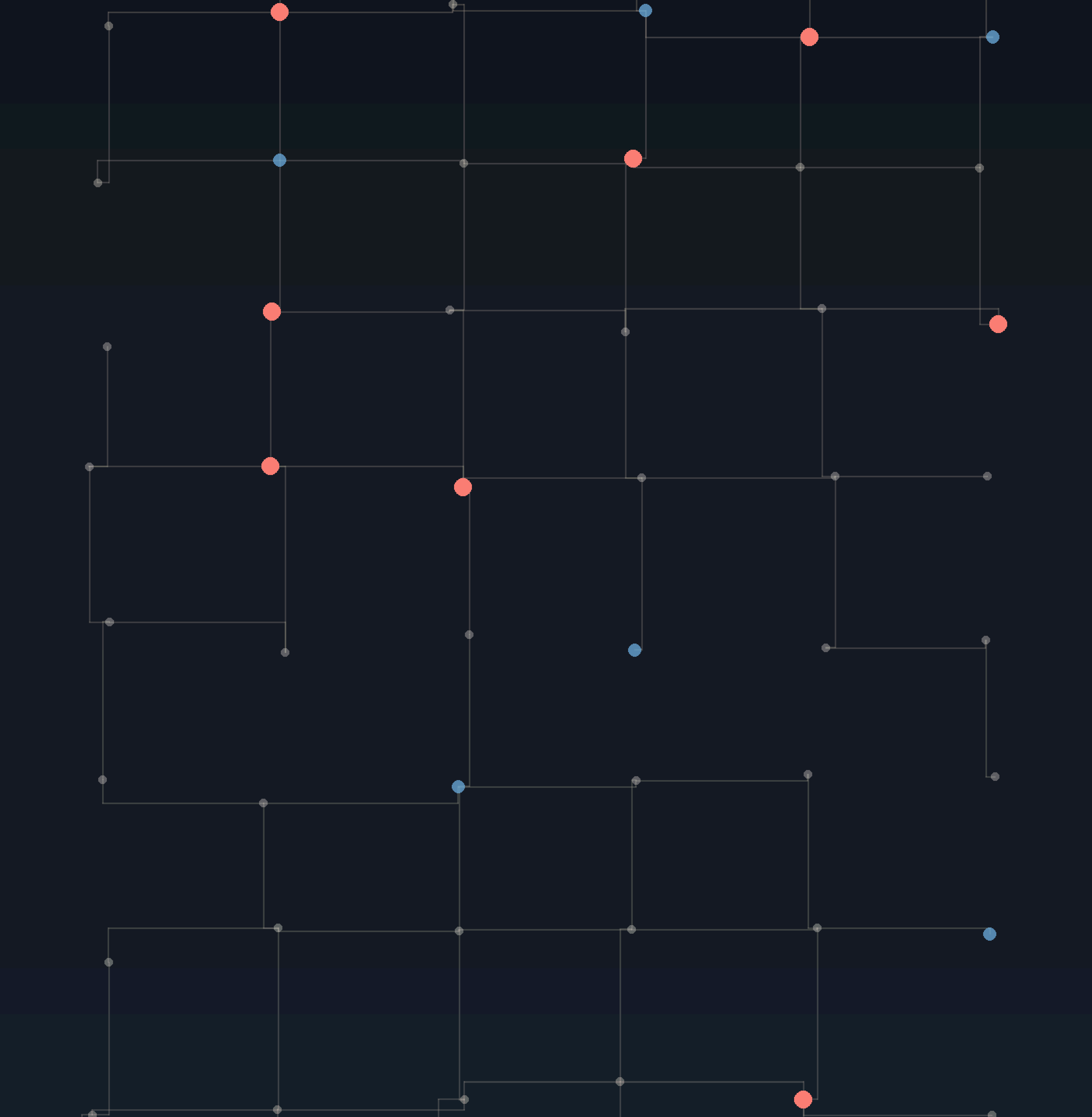
(a) State the front door's action for each.
(b) One careless colleague proposes "just
return the nearest cached answer,
always." **Name the doctrine that
forbids it, and the analogy that made
it famous.**

Serve, hint, miss — never guess

(a) $0.95 > 0.92$: **serve** the cached answer.
 $0.80 < 0.86 < 0.92$: **do not serve** — pass
downstream as a retrieval hint. $0.74 < 0.80$:
miss; run the full cascade.

(b) **Never KNN without a threshold.**

The famous collision: type-1 and type-2 diabetes — nearest neighbors, clinically opposite answers. Every team that thinks the semantic cache is easy builds one, annoys its users, and rips it out. Built with E and τ treated seriously, it is magic.

An abstract graphic on the left side of the slide. It features a dark blue background with a light blue grid of thin lines. At the intersections of these lines, there are small dots. Some dots are colored red, some are blue, and some are grey. The dots are scattered across the grid, with a higher concentration of red dots in the upper left and lower right areas, and blue dots in the upper right and lower left areas.

MILESTONE · ACT VI OF VII

The Machine Room

Recall is bought with milliseconds, storage is a budget line, and the index is a living thing.

Recall is bought with milliseconds

One billion vectors, one HNSW-class engine, one dial: **70% recall at 1.7 ms p99** — or **98% recall at 12 ms p99**. Same index, same hardware; the difference is how hard the graph is searched.

DiskANN puts billions on SSD; CAGRA puts the graph on the GPU.

*"What recall does the ANN index give?" is a malformed question. The honest form: **what recall, at what latency, at what scale** — an exchange rate your latency budget must sign, per lane, at p99.*

Quantization, down to one bit

RaBitQ made binary respectable — with a theoretical error bound; Elastic shipped it as BBQ. Voyage's **binary vectors at 512 dims** run **~200× less storage** than full-precision frontier embeddings — at **better** retrieval quality than full-precision OpenAI.

*The pattern that ships: search the **binary or int8** index, take a generous candidate set, **rescore the survivors** at full precision — the scoop-and-sieve principle, applied to bits. Precision is a budget line now.*

A query must never span embedder versions

Embeddings from different models — or different fine-tunes of the same model — live in **different coordinate systems**. Cosine between them is numerology.

Stamp every vector with `model_version`; make the query path **assert** it.

*Version skew is the machine room's silent recall killer: nothing errors, nothing logs, the geometry is simply wrong. The 13-month corpus autopsy that found **89.7%** of live chunks failing an **integrity check** — mostly stale versions — is what unattended indices become.*

The dual-write migration playbook

- (1) Provision generation $N+1$ alongside the old.
- (2) **Dual-write** all live changes to both. (3) Backfill history through the new embedder. (4) Shadow-compare retrieval overlap. (5) Flip the **collection alias**. (6) Keep dual-writes and a snapshot until rollback is unthinkable.

*An embedder upgrade **is** an index migration — Act II's 48-hour fine-tune ends here, behind an alias flip gated by the yardstick. Notion runs this as generation-based rollover with sub-minute freshness; span-hashing cut writes 70%.*

Poisoning — five documents in millions

PoisonedRAG: injecting **5 crafted documents** into a corpus of millions steers ~**90%** of targeted queries — far below any statistical radar. And LLM-polluted corpora collapse quietly: at 50% contamination, 68–76% of retrieved evidence is synthetic, with accuracy flat and **no alarms**.

*The retrieval layer is an attack surface, not plumbing. Watch the **archaeology, not the accuracy** — provenance telemetry is your intrusion detection.*

(Guardrailscraft handles the injection payloads themselves.)

ACL pre-filtering, never post-filtering

Sync permissions into payload metadata **at index time**; filter **inside the ANN search** at query time. Post-filtering under-returns — and leaks existence through ranking and latency side-channels. Per-user namespaces do not scale past a pilot.

*A hard security boundary belongs in the query, not in a polite cleanup afterward. Filtered ANN is a solved engineering problem in every serious engine; use it — the collection is shared, the **rows** are not.*

The cost ledger

The marginal query: embedding is micro-cents; ANN reads are cheap; the cross-encoder dominates — **~\$0.002–0.005 per query, rerank-dominated**. Generation then costs 5–50× the whole cascade.

The standing costs: the multi-index portfolio's storage multipliers, and re-derivation of artifacts as the corpus churns.

*Divide before you compare: the CFO's metric is **dollars per correct answer**, and the cascade is the cheapest line on that invoice.*

The expensive lines are the generator — and the mistakes.

Quiz I I — the suspiciously cheap terabyte

A vendor claims binary quantization gives you "200× storage, free." Your baseline: 3072-dim float32 vectors (12,288 bytes each). Binary at 512 dims: 64 bytes.

(a) Verify the 200× arithmetic. (b) What word in the pitch is false, and what two-step serving pattern makes the claim almost true?

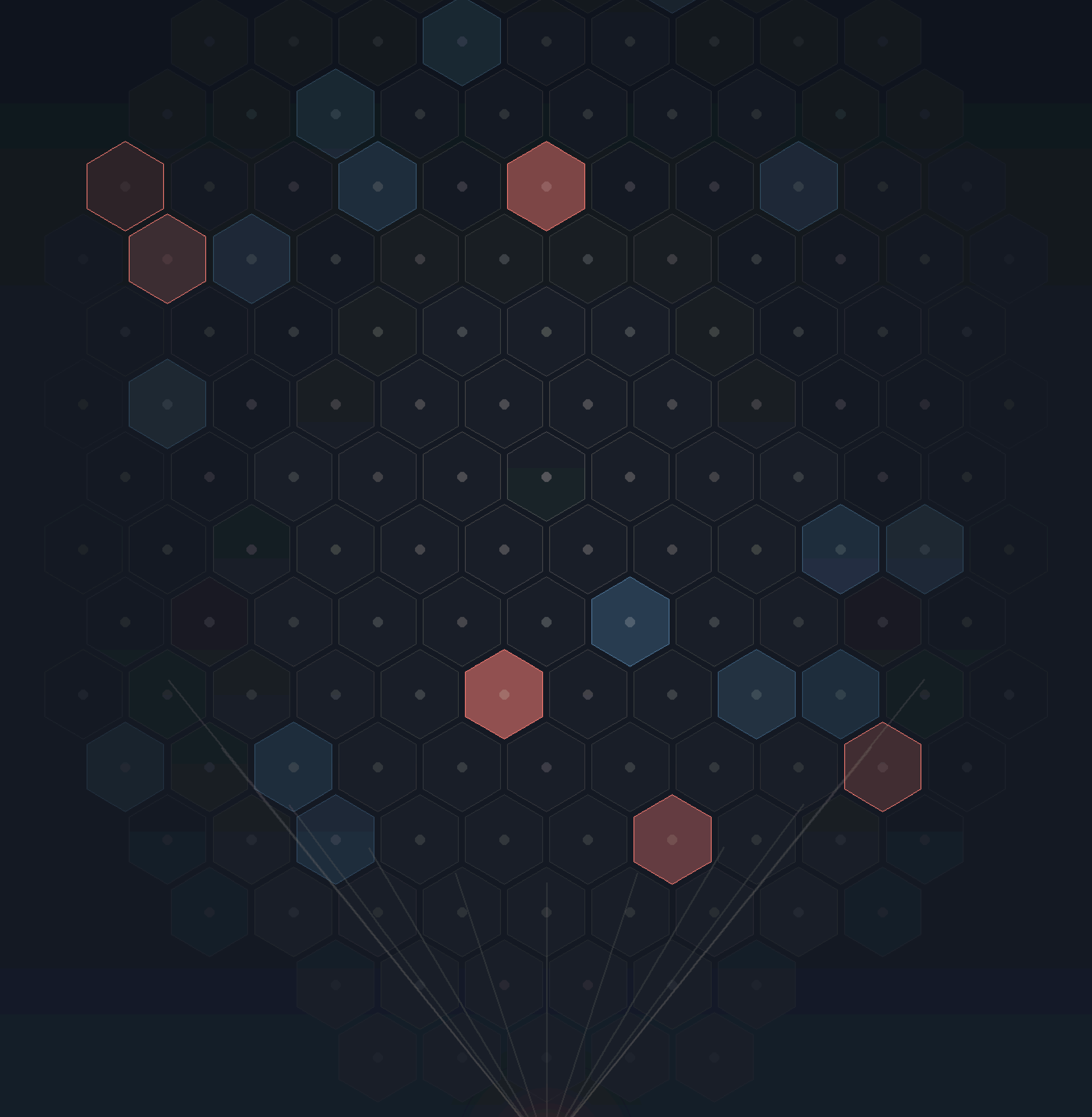
Say back the machine-room refrain.

The arithmetic holds; the "free" does not

(a) $12,288/64 = 192\times$ — the $\sim 200\times$ is honest arithmetic.

(b) "Free." Bits discard geometry; recall must be **re-measured**, and the shipping pattern is: search binary, take a generous candidate set, **rescore survivors at full precision**.

*Scoop and sieve, again — this time in bits. And the refrain: **recall is bought with milliseconds** (and storage, and precision); every cheapening of the scoop must prove, on the yardstick, that it did not sell the marbles.*



MILESTONE · ACT VII OF VII

The Seeing Catalogue

The corpus was never only text. The portfolio principle, extended to pixels.

ACT VII · THE LANE THAT CANNOT READ

CLIP sees style, not sentences

Joint-embedding two-towers (CLIP → SigLIP → SigLIP 2) are superb at appearance — and structurally blind to composition: **Winoground** ~8% vs 85.5% human; relations near chance; an effective text window under **20 tokens**; no reading of text inside images.

*SigLIP is the coupling column's epilogue: the **sigmoid loss** decouples pairs the way DCL decouples the two forces. Your Part III conviction, wearing vision clothes.*

THE LIBRARY OF MANY CATALOGUES · SUPPORTVECTORS

The caption lane

A VLM description, embedded as text, image in the payload — retrieve the derivative, return the source, for pixels. Otlet ran this lane by hand: image repertories indexed by cards.

One caption is **one projection**. Take several: dense caption · **OCR lane** · attributes · synthetic visual Q&A.

Fusing the visual lane with the caption lane by RRF tops every published single-lane result. Two angles, one reconstruction — tomography again.

ColPali, and Chandar's lab

Treat the **page as pixels**: a VLM emits patch embeddings; query tokens **MaxSim** against patches. ViDoRe's founding result: **81.3** nDCG@5 vs 65–75 for OCR and caption pipelines.

The lab's ColQwen module is this lane. ViDoRe v3 says the problem is **not** solved: best models $\approx$ **65%**.

*No OCR, no chunking, no caption
— the document **is** the
representation. Late interaction's
rare-term magic, now for layouts,
stamps, and tables.*

ACT VII · THE TAX, REPEALED

The storage bill of seeing

Patch multivectors cost up to $\sim 2,600\times$ a bi-encoder. The repeal: binary quantization ($32\times$, nearly free after rerank), row pooling ($13\times$ at 0.95 retention), **MUVERA**.

Worked: 1M pages — **393 GB** float32 $\rightarrow$ **12.3 GB** binary $\rightarrow$ **4.1 GB** pooled: parity with a single-vector index.

Recall is bought with milliseconds; seeing is bought with bytes — and the bytes just went on sale. The cost objection to ColQwen fell in 2025, same as ColBERT's.

The unified embedders

LLM-backed multimodal embedders (Voyage, Cohere Embed v4, Jina v4, GME) put a real language model behind the text side. voyage-multimodal-3.5 sits **within 0.29%** of the best pure-text embedder — the multimodal **tax** on text is gone.

Jina v4's own multi-vector mode beats its single-vector mode by **7–10%** on visual tasks.

*The premium is now a **per-query routing decision**, not a model choice. Vendor-measured numbers — weigh accordingly, then ablate on your yardstick.*

The exhibit joins the docket

No second architecture: image lanes **fuse by RRF** into the same court — parent-dedup by image id, cheap **MaxSim rerank** as the free lunch, VLM rerankers (7–14 s) for offline and high-stakes only.

And before any lane earns its place: **50–200 real image queries, graded**. The yardstick, first.

The compound eye is the architecture: many ommatidia, each nearly blind, one fused visual field. The exhibit enters the courtroom; the law does not change.

Quiz 12 — the drawings that wouldn't be found

A client has **1M scanned engineering drawings** — title blocks, part numbers, revision stamps. Their CLIP index retrieves "hydraulic manifold, rev C" hopelessly.

(a) Name the two lanes you add, and why CLIP alone was structurally doomed here.

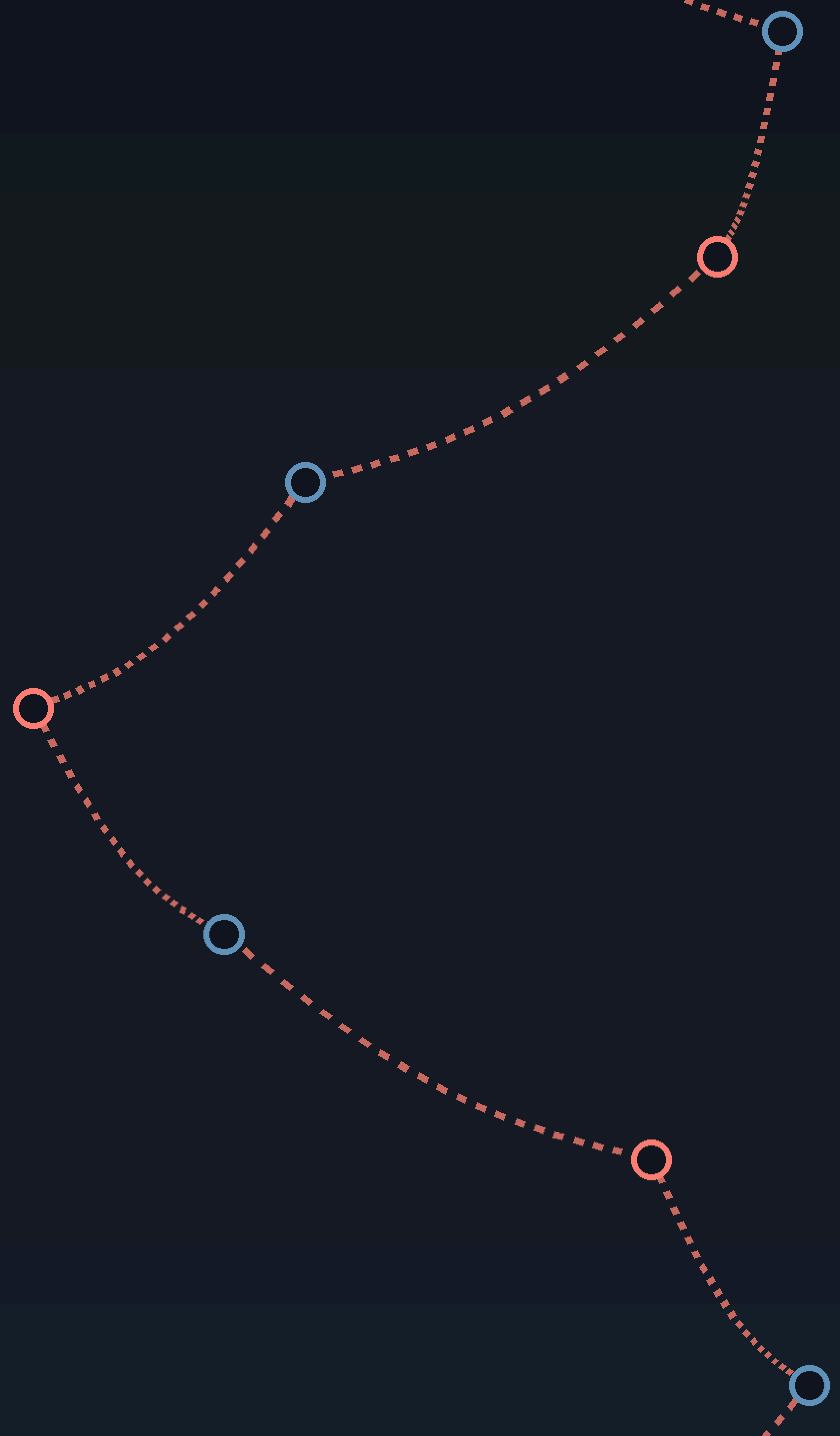
(b) Compute the ColQwen bill at float32 (~393 GB/M pages) after binary quantization and row pooling. Then say in one sentence why the OCR-free lane and the OCR lane are not redundant.

Quiz 12 — answered

(a) ColQwen lane (patch MaxSim reads title blocks and stamps *in situ*) + an OCR text lane (part numbers are exact-match, SPLADE territory). CLIP: composition-blind, ~20 effective tokens, cannot read.

(b) 393 GB → **12.3 GB** binary (32×) → **4.1 GB** pooled (~96×): single-vector parity.

Say-back: the OCR lane retrieves what the page **says**; the patch lane retrieves what the page **is** — layout, stamps, geometry — and their failures are uncorrelated, which is the only reason to keep two of anything.



CODA · THE WHOLE OF IT, AS
ONE DISCIPLINE

The Playbook

Scale decides architecture.
Not taste. Not fashion. Scale.

The square-cube law of retrieval

Double a creature's size: surface grows as the square, weight as the cube — the insect's **exoskeleton** physics stops working, and the body plan must be reinvented. There is no perfect architecture; **only an architecture perfect for a scale.**

10K docs: BM25, a big context, one reranker — worn proudly. 10M heterogeneous docs: the full portfolio and court.

*The one-document demo is correct engineering at that scale. So is the twelve-index cascade at its own. The recurring question is never "what is the best architecture?" but "what is the best architecture **at this scale?**"*

Every index must earn its place

Climb one rung at a time — baseline first, and the baseline is sacred. Gate every rung with the yardstick; estimate each component's **Shapley-style marginal contribution**; remove what removal does not hurt.

Two named pathologies: the **Kitchen Sink** (every arXiv technique, adopted) and the **Sacred Cow** (the component kept for sentiment). And above every ring, the newest rung: the **governed concept layer** — earned only for the governed core.

Every index must earn its place through ablation. The pipeline without ablation is not a system — it is a superstition. The storage bill is decided by the metrics, not by the engineer's enthusiasm for lenses.

The reference architecture

- **Front door** — semantic cache (E, τ) on the canonized query; transformation cascade; router by altitude and modality
- **The portfolio** — SPLADE · Matryoshka-128 scoop · full-dim rescore · ColBERT; derivative indices: factoids, QA pairs, rewrites, summaries, RAPTOR nodes
- **The court** — RRF fusion → parent-level dedup → MMR → cross-encoder on source text → calibrated verdicts, 20–30 seats
- **Sidecars** — GraphRAG for the sensemaking 1–2%; text-to-SQL for schema questions
- **Two closed circuits** — the **training loop** (production negatives → curriculum → DCL fine-tune → migration) and the **eval loop** (gold set, recall@scoop per lane, NDCG+recall pair, archaeology telemetry, drift watch)

One corpus, many catalogues, one court — drawn as a single map. Everything on it earned its place through ablation, and everything on it is watched.

Every component posts twice

Luca Pacioli did not invent double-entry bookkeeping — he **codified** it, and Venice ran on it.

The retrieval ledger: every component has **two columns** — what it costs to run, and what its **absence** costs you. A component with no measured entry in the second column is decoration.

*For each of the 18 components: **[To adopt]** the practical steps · **[Without it]** which query class degrades, which metric shows it · **[Verdict]** at small, mid, large scale.*

The estimator ladder

Leave-one-out **lies** when components entangle:
substitutes **share** credit (dedup × MMR),
complements **hoard** it (factoids × QA).

Exact Shapley needs $2^{14} = 16,384$ ablation
builds. The practical rung: **truncated Monte-
Carlo permutation sampling**, plus archaeology-
of-top- K win-share as the free, always-on
proxy.

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

*Win-share is presence, not
causation — the judges never
appear in top- K at all. Trust it as a
pointer, audit it with the
estimator.*

Quiz 13 — the unallocated residue

Four components. Leave-one-out
deltas: factoids **.05**, QA **.05**, dedup **.04**,
cross-encoder **.12**. Sum: **.26**. But
removing all four costs **.28** of NDCG.

*Where did the missing **.02** go? Then say
in one sentence why leave-one-out cannot
be the whole truth for a portfolio whose
components substitute and complement
each other.*

Quiz 13 — answered

The .02 is **interaction value** — earned by pairs, credited to no one. LOO measures each component against the **full** coalition of the others; substitutes split their shared credit to near-zero, complements pocket each other's help.

Shapley: $\sum_i \phi_i = v(N) - v(\emptyset) = .28$ — every point allocated.

Say-back: leave-one-out prices a component only at one margin; Shapley averages its arrival over every possible order, so the entangled credit has nowhere to hide.

Databases ate the model layer

MongoDB bought **Voyage** (~\$220M, Feb 2025). Elastic bought **Jina** (Oct 2025). OpenAI bought **Rockset**; Anthropic runs on **Turbopuffer** — 22 people, object-storage economics, trillions of vectors.

Every managed RAG SKU now ships the same disclosed cascade: hybrid → RRF → rerank → query decomposition. **The architecture you built by conviction became the industry's convergent default.**

The standalone vector-DB era is over; retrieval models are becoming database features. Verify any row of this landscape before quoting it — no chapter rots faster than the wilds.

Own the law; rent the courtroom

Own: the yardstick, the fine-tuned embedder, the fusion weights and depths, the provenance graph — everything that encodes **your** corpus and **your** judgment.

Rent: ANN serving, rerank APIs, the web-search lane — courtrooms are commodities; the law is not.

Otlet's Mundaneum stood in a borrowed palace. The tragedy was never the eviction — it was losing the catalogue. Rent the building; never the repertory.

Quiz 14 — the stories, reconstructed

*Before the map is unveiled: the room
reconstructs the day from its stories.*

Which story taught which doctrine?

*(a) the marbles in the sandbox (b) the
classmate's forty questions (c) the \$1,000
intern invoice (d) the scoreboard nobody
could normalize (e) the decoy on the
witness stand (f) Galileo's giants*

*Say each doctrine back in its own
words — no notes.*

Six stories, six load-bearing walls

(a) the recall ceiling — the scoop decides what is caught. (b) the user is the examiner — manufacture the derivative corpus. (c) the sidecar doctrine — telescopes are not for wristwatches. (d) fuse in ranks — RRF over incommensurable scores.

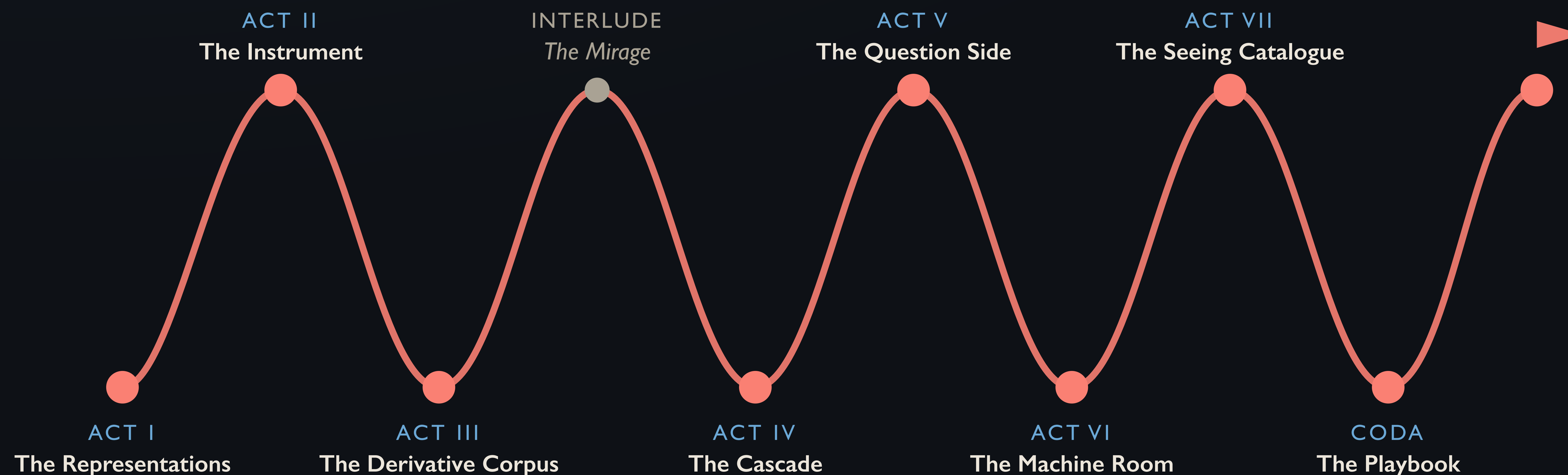
(e) the decoy's job ends at retrieval — rerank the source.

(f) scale decides architecture — square-cube, exoskeleton to endoskeleton.

*And behind all six, Otlet's Mundaneum: **one corpus, many catalogues, and a court to adjudicate among them.** Now — the map, traveled.*

PROMISES KEPT · THE SAME MAP, TRAVELED

The journey, behind us



The WHY

Retrieval is the **load-bearing wall** of RAG: generation cannot cite what retrieval never surfaced, and recall at the scoop is a ceiling no downstream virtue raises. The register gap — queries are questions, corpora are prose — is permanent.

So one index was never going to be enough.

*A retrieval system is a **portfolio of representations** plus a **court of adjudication**. Architecture is choosing the portfolio and tuning the court; **scale decides how much of each you can afford**.*

The WHAT

The portfolio — SPLADE, Matryoshka dense at two widths, ColBERT; the derivative corpus (factoids, QA pairs, rewrites, summaries, RAPTOR); the GraphRAG sidecar. **The court** — RRF at $k=60$, parent-level dedup, MMR, a calibrated cross-encoder on source text.

*And the two flanks: the **trained instrument** (DCL, the curriculum, the MRL invariant) and the **question side** (six pathologies, the cache's E and τ , routing, the agentic residue). The machine room keeps it honest at p99.*

The HOW

- **Yardstick first** — no index without a gold set to earn it on
- **Fine-tune in 48 hours** — curriculum negatives, DCL at every Matryoshka prefix, regression-test the 128-d scoop
- **Fuse in ranks; dedup by parent; rerank the source**
- **Route by altitude** — sidecars for sensemaking and schemas
- **Pre-filter ACLs; stamp versions; watch the archaeology**
- **Escalate by ablation** — scale, not fashion, decides

*Scoop wide and sieve hard,
manufacture what the examiner
will ask,
let cheap judges spend their
pennies
so the supreme court can afford its
verdict.*

Each a load-bearing wall

- **Formal et al. (2021), SPLADE** — the sparse lane that sees what the page means
- **Kusupati et al. (2022), Matryoshka Representation Learning** — dimensionality becomes a runtime cost knob
- **Khattab & Zaharia (2020), ColBERT** (+ Santhanam 2022, ColBERTv2) — late interaction, and the compression that saved it
- **Yeh et al. (2022), Decoupled Contrastive Learning** — the positive leaves the denominator
- **Chen et al. (2023), Dense X Retrieval** — propositions as the retrieval unit; Otlet mechanized
- **Sarathi et al. (2024), RAPTOR** and **Edge et al. (2024), GraphRAG** — the zoom lens and the telescope
- **Cormack, Clarke & Büttcher (2009), Reciprocal Rank Fusion** — three authors, one page, $k = 60$
- **Modarressi et al. (2025), NoLiMa** and **Zou et al. (2024), PoisonedRAG** — the mirage measured; the adversary counted

The card outlived the building

The surviving drawers stand on the rue de Nimy in Mons. A catalogue is a **discipline, not a place** — and this summer the card came back, as typed markdown with provenance in its frontmatter.

1940 was irreversible because the pipeline was clerks and forty years. Ours are derivatives — burn our catalogues and we re-derive them.

The Mundaneum, Mons. (Mr N, CC BY-SA 4.0.)



Thank you

The occupiers emptied Otlet's halls in 1940; tonnes of the catalogues were destroyed, and the representations of a lifetime could not be rebuilt — the pipeline was human hands.

Ours can be. Every catalogue we built today is a derivative, regenerable from the source. The corpus is the only irreplaceable artifact. Guard it, project it many ways, let the court decide — and prize, above every index, the architecture that knows its own scale.